

Ryan Barkman

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Portfolio



LinkedIn



SUMMARY OF QUALIFICATIONS

- Skilled with **Unity** and **Unreal** game engines for developing small 3D experiences
- Strong **Unity** knowledge of **C# scripting**, **finite state machines**, and **nav meshes**
- Proficient in multiple programming languages, including **C#**, **C++**, and **Python**, with a strong focus on game development
- Actively expanding knowledge of Unity's **mobile features** and their **version control** through a card game development project
- **Collaborative team player** with a proven ability to manage individual tasks while contributing to group discussions, offering feedback, and keeping team members informed of progress and challenges
- Currently developing a **large-scale 2D game** in the early requirements phase, with an understanding of the full development lifecycle
- Committed to **version control** through platforms such as **Bitbucket**, **GitHub**, and Unity's built-in version control system
- Experience managing custom game engines and graphical engines utilizing **SDL** and **OpenGL**
- Enjoy playing and testing video games including my own, with a deep understanding of gameplay systems and user experiences

TECHNICAL SKILLS

- Expertise in **C#**, **C++**, **Python**, **HTML**, and **SQL**, with a good understanding of each language's strengths and applications
- Strong use of **debugging** and using the **call stack**
- Experienced in a variety of development platforms, including **Visual Studio**, **Unity**, **Unreal Engine**, and **Visual Paradigm**
- Strong understanding of web-based tools and technologies, including MVC architecture, APIs, data validation, helper tags, authentication, and Razor Pages
- Detail-oriented tester with a keen eye for identifying and troubleshooting errors in both game mechanics and software functionality

EDUCATION

Honours Bachelor of Computer Science; Specialization in Game Engineering

Sept 2022 – Present

Sheridan College, Oakville ON; (4-year program)

- GPA: 3.68
- Relevant Coursework:
 - **Statistics for Data Science** (81%) - Python-based data visualization using Pandas, NumPy, and Matplotlib, statistical calculations
 - **Game Engine Architecture** (93%) - Created a game engine from a template to calculate game calculations and display them in C++.
 - **Computer Graphics and Anim.** (96%) - Created a graphical engine from a template to render game objects from blender as well as shaders in C++.
 - **Artificial Int. for Games** (81%) - Used unity to learn and implement NPC logic. Choice decisions, Navigational Movement, and finite state decision trees.
 - **Multiplayer Games** (100 %) - Used unity to learn and implement multiplayer principles. Voice chat, Server and Client control, and data passing.

- **Capstone Inception** (85%) - Led initial project planning and elaboration setup. includes user story mapping in Jira, sprint planning, and defining use cases in Visual Paradigm.
- **Capstone Elaboration** (90%) - use inceptions to plan and turn concepts into a physical product. Managed sprints and scrum meetings to ensure task completion.

PROJECTS

Capstone - Oregon's Fall

- Pitched and led the design and development of Oregon's Fall, managing project scope and team deliverables across three course phases (Idea planning, use cases, build from use cases).
- Facilitated weekly Scrum meetings, sprint planning, and task allocation to maintain alignment and deadlines.
- Implemented core gameplay mechanics including travel logic, custom binary save/load systems, and modding concepts.

Tanks

- Created a 2-player multiplayer demo featuring simple real-time environmental destruction and projectile trajectory.
- Implemented lobby and room management to handle state synchronization and player connections.
- Used server-side control managing the two clients.

Planet Conqueror

- Created a strategy game featuring an 'AI' opponent driven by a finite state machine for real-time planet acquisition.
- Impossible for the player to win as the computer takes positive moves as soon as they are available.

PERSONAL PROJECTS

Physical Board Game

- Designed a board game focused on island movement from card auctions. Currently thinking that it could be turned into a digital adaptation.

Cryptic Crosswords

- Constructed a cryptic crossword puzzle with wordplay clues, inspired by Minute Cryptic.
- Example: "Coal Pit "E" strangely claimed five lives. (3)(5)" → Ace Pilot (Anagram of "Coal Pit E" meaning someone who has "claimed five lives").

Creative Writing

- Goal is to write a full book every year; current book at 15,000 words.

WORK EXPERIENCE

Kitchen & Stock Clerk

Aug 2022 – Sep 2024

McDonald's

- Maintained accuracy and speed while managing food items in a fast-paced, high-demand environment, contributing to efficient workflow and customer satisfaction
- Facilitated clear communication across the team to helping sustain high service standards
- Demonstrated excellent attention to detail when managing new stock and product

Garden Center Associate

Jun 2022 – Aug 2022

Battaglia's Marketplace

- Demonstrated excellent customer service when assisting customers with product selection